1. MOC type – Usually denoted as $[M_xL_y]^n$[counter_ion]$^{-n}$, whereM is the metal ion, L is the ligand, and counter_ion is the species used for charge balancing the complex. Given the high complexity of L, most papers would just show it or name it once, then refer to the MOC according to the formula above or by 'MOC 1'/ 'Cage 1'. That is fine, it's quite common in the community. Note that the stoichiometry is essential, as it influences the geometry of the complex, a defining feature.
2. Reactants – Metal ion source (usually a salt), ligand L, and any other sources of counter_ion that, when mixed together, will self-assemble as a MOC
3. Reactants Synthesis – some ligands L can be commercially purchased, or need to be pre-synthesised, shown in this step.
4. Process Parameters – the actions (steps) required throughout the synthesis of the MOC (e.g. M was mixed with L at X degrees C, followed by removal of water in vacuo).
5. Characterisation – how the product MOC was characterized (e.g. NMR, mass-spec, PXRD, IR, DOSY, etc.)
6. Host guest properties – can be none, but most papers will include an application of the MOC that revolves around trapping different small molecules (guests).
7. Others – e.g., stability, other properties measured for the cage (such as but not limited to, conductivity, BET adsorption, other type of characterization), any stereochemistry.